

Solution Requirements

Date	2 JULY 2026
Team ID	XXXXXX
Project Name	OPTICROP – SMART AGRICULTURAL PRODUCTION OPTIMIZATION ENGINE
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can access the crop recommendation system through the web interface and submit agricultural parameters for prediction.	High
2	Authorization levels	Farmers can enter data and receive recommendations. Administrators can manage datasets and system updates.	Medium

3	External interfaces	System interacts with the trained Machine Learning model, dataset files, and Flask web application interface.	High
4	Transactions processing	Soil and weather data entered by users are processed and sent to the ML model to generate crop recommendations.	High
5	Reporting	System displays recommended crop results and prediction outputs through a dashboard.	Medium
6	Business rules, etc.	Crop recommendation is generated only when valid values for N, P, K, temperature, humidity, pH, and rainfall are provided.	High
7	Compliance to laws or regulations	User information is used only for prediction purposes and follows basic data privacy practices.	Low

8	<i>Other</i>	Future enhancements include fertilizer recommendation and real-time weather integration.	Low
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Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Prediction results should be generated quickly after user submission.	Response time < 2 seconds
2	Scalability	System should support additional datasets and multiple user requests.	Supports 100+ concurrent users
3	Security & Data Privacy	User input data should be protected from unauthorized access.	Secure data handling
4	Reliability & Availability	System should remain operational during usage and testing.	99% availability

5	Usability & Accessibility	Interface should be simple and easy to use for farmers and researchers.	User-friendly responsive design
6	<i>Other</i>	Maintainability and portability across different operating systems.	Runs on Windows, Linux, and macOS